


## Objects swapping positions

Moving in the world frame: **the man**, **the woman**. Staying in place: the tank, the tree, the fountain, 2 chain barriers, the lamp post.

**The man** moves about 0.5 m in the world frame at an average speed of 0.23 m/s; it reverses direction around frame 15; it approaches the woman.

**The woman** moves about 0.3 m in the world frame at an average speed of 0.26 m/s; it accelerates over the clip; it reverses direction around frame 21; it approaches the man.

### Question

**Right answer: B**

In the world frame (ignoring camera motion), what was the average speed (in meters per second) of **the woman** between frame 25 and frame 32 of 34?

A. 0.21 **B. 0.39** C. 0.52 D. 0.29

reference: 0.43 m/s

### Qwen3-VL-8B · frames only

```
<evidence>
object: the woman
position_frame_25_m: (-0.8, 0.4, 3.0)
position_frame_32_m: (-0.4, 0.5, 3.1)
displacement_m: 0.4
time_s: 1.40
speed_mps: 0.29
</evidence>
<answer>D</answer>
```

### Ours · frames + trajectory tokens

```
<evidence>
object: the woman
position_frame_25_m: (-0.9, 0.2, 2.4)
position_frame_32_m: (-0.4, 0.3, 2.7)
displacement_m: 0.6
time_s: 1.40
speed_mps: 0.40
</evidence>
<answer>B</answer>
```